SMA-1 Executive Report

Date: 2026-06-12 · **Status:** research prototype with held-out, multi-seed headline results · **Repo state:** all 11 test gates green, full history committed, ontology hash-frozen at tag `ontology-v1`

What SMA-1 is, in plain words

A memory system for AI that remembers *how things happened* (the causal shape of an incident) instead of *what words were used*. Raw data enters memory only through deterministic rule-based parsers — no AI in the extraction path, so the memory's contents are bit-for-bit reproducible and auditable. An LLM sits strictly downstream and can only verbalize what was retrieved, with provenance on every claim. The committed product wedge: **infrastructure incidents** — "have we seen this before, what caused it, what fixed it."

The four headline results

1. Structure transfers across systems; word-matching does not (the H1 result)

We froze the event ontology (git tag, hash-pinned), then downloaded a supercomputer log corpus nobody had touched (Spirit) and asked: can memory built from one system (BGL) answer queries from the other?

BGL → Spirit (3 seeds, frozen ontology)	macro-F1
SMA	**0.92 / 0.965 / 0.93 (mean 0.938)**
WL graph-similarity (same representation)	0.62
Production RAG (hybrid + reranker)	0.59 / 0.52
Dense embeddings / BM25 / entity-graph	0.31–0.41

Not post-hoc (ontology frozen first), not single-seed, not vocabulary-assisted. The decomposition is clean: our representation alone gets generic graph similarity to 0.62; the structure-mapping alignment adds the remaining +31 points. **Both halves of the system earn their keep, and production RAG does not transfer.**

Honest scope: transfer holds *within* a failure-physics family (infrastructure logs). Cross-family pairs (HDFS→OpenStack, HDFS→Spirit) fail for every method — a real boundary, documented, with a research idea attached (missing-event anomalies via schema expectation-violation).

2. Within one system, SMA beats the full production-RAG ladder where structure matters

HDFS triage (anomalies live in event *patterns*): SMA 0.955 > Hybrid-RRF 0.835 > frontier-LLM long-context 0.809 > Hybrid+Rerank 0.743 > BGE 0.638 > SPLADE 0.404. And on the deeper metric — retrieving the correct *failure family*, not just "an anomaly" — SMA leads at 0.906 vs BM25's 0.62.

Honest counterweights: on BGL, where anomalous messages literally announce themselves ("KERNEL FATAL"), lexical methods saturate (~1.0) and SMA trails — and a cheap WL graph-similarity pass actually beats full structure-mapping within-system (0.98 vs 0.955 at 1/30th the latency). Design implication adopted: tiered retrieval (cheap graph pass within-system, full mapping for cross-system and provenance).

3. Provenance discipline measurably prevents confabulation (the H3 result)

200-cell study (20 questions, half unanswerable/false-premise, 5 memory modes, 2 LLMs), independently judged with an audit trail: the provenance-bound DeepSeek verbalizer was **99% correct with zero invented entities** and abstained on 50/50 unanswerable questions; an undisciplined small model confabulated 18% of the time and never abstained once. Honesty is enforced by architecture + a competent verbalizer — and because evidence is structured, the *unsupported-claim rate* is now a measurable, reportable metric (~0.02 claims/answer vs 0.32).

4. The engineering holds up

~2000x matcher speedup (worst case 5 min → 181 ms); certified retrieval bounds fixed; three label-leakage vectors found and closed across datasets; sampler nondeterminism fixed; every fix gated by 11 green tests and an append-only ledger. $0 LLM tokens for extraction/indexing; CPU-only.

What this adds up to

SMA is not a RAG killer — within-domain document QA over prose remains RAG's territory, and our own BGL numbers prove surface methods win when vocabulary carries the signal. SMA is the first credible **post-RAG memory architecture for event-structured data**: it wins precisely where RAG structurally cannot follow (vocabulary shift, cross-system reuse, auditability) and degrades honestly where it shouldn't be used. The Fortune-10 objections (post-hoc ontology, weak baselines, shallow metric, verbalizer-vs-memory safety) have each been answered with a measured experiment rather than an argument.

Blueprint completion (honest)

- P0–P6 (system build): complete at fixture-gate level; G2's 25-pair SME-v4 oracle battery and G3's LogHub-2k template validation remain for full rigor.
- P7 (evaluation): within-system, scorer ablation, held-out transfer, family metric, H3 with judging — done. Remaining: SSB de-circularization, calibration→freeze→prereg tag, multi-seed statistics with bootstrap/CIs, ablation battery (γ=0 first), drift protocol T5 (the "agentic" claim), BugsInPy (T3), ARN (T4).

- P8 (paper/artifact): not started; Docker repro and writing remain.

Next steps (in order)

1. Rare-family-stratified scorer analysis, MDL corpus-cost upgrade, ses_n bias study → resolve the scorer default → calibration freeze + prereg tag.
2. Single-shot test runs with paired-bootstrap statistics (numbers → claims).
3. Drift protocol T5: wrong-action rate under concept drift (the result that would carry a Nature-tier claim).
4. Latency ladder to p95 < 300 ms @ 100k cases (real HNSW, parallel FAC, tiered WL/SME retrieval, Rust hot loops).
5. Second domain adapter end-to-end (pharmacovigilance design exists) + paper writing + open artifact.

Where everything lives

`reports/report.html` (full narrative + every table) · `reports/*.csv` (raw artifacts incl. h3_judged.csv awaiting your spot-check of low-confidence rows) · `docs/STATUS.md` (append-only history) · `docs/ADR/` (decision records) · `docs/POSITIONING.md` (the wedge) · git tags: `ontology-v1`.